Rho step cost across isogenous curves of $\mathbb{F}_{2^{31}-1}$: a curve-level effect, confounded with execution order

Phase 4 of `research/fp31_whole_field_census_20261111`, 2026-11-12. Protocol `phase4/PROTOCOL.md`, frozen before data (sha256 `cd2984ea…`). Card `CARD-fp31-phase4-rhostep-20261112-1`. The director assessed the execution as *impeded*: 45 of 48 classes completed. Every claim here is **[pcc:unverified]**.

What ran

- **Population.** 48 prime-order classes, each with its isogenous curves plus 2 isomorphic models.
- **Design.** The same planted k on every curve (`planted_shared`); 64 targets × 2 rounds per curve.
- **Arms.** `rho.plain`, a `rho.plain` A/A arm, and `rho.negation` as the positive control.
- **Metric.** Wall ns per counted group operation (GAE).
- **Volume.** 150,144 measured runs, every answer correct. 12,283 rows were graded below L2.
- **Missing classes.** c046, c047 and c048 were refused at start: the host was not quiet because of a concurrent build in this session. The rows of c052 were graded L0 because shell and Python processes of this session ran on the reserved CPU.

Frozen result: effect suspected (PROTOCOL §6)

- **T1.** Within-class Friedman p-values across curves are not uniform: KS p = 3.5×10^{-15}. 10 of the 45 classes are below 0.05/45.
- **Size.** The between-curve SD of log ns/GAE is 0.95 %, with a 95 % upper bound of 1.07 %. The per-class max/min ratio has median 1.038 and maximum 1.081.
- **Controls.** The A/A arm reads 1.0009 ± 0.0005. The positive control (`rho.negation`) reads 1.094, so the detector works.

Why the effect is not evidence of cheaper arithmetic (exploratory diagnostics, after the frozen analysis)

- **Execution order.** ecbench runs workloads in spec order within each round; `order: shuffle` permutes only the arms. Each curve's runs therefore fill one contiguous time window (9 curve switches in 1,920 runs of class c039). The curve is confounded with time.
- **Persistence.** Per-curve speed persists in the A/A arm (mean rank correlation 0.64, 43/45 positive) and in the negation arm (0.59, 43/45). The A/A arm replays the identical walks (same algorithm seed and the same GAE count in all 1,280 pairs). The negation arm takes different walks.
- **Isomorphic models.** An isomorphic model (the same group, other a, b) differs from its base curve as much as unrelated curves do: median $|\Delta \log|$ 1.2 % against 1.3 %; probability that the iso pair is closer 0.49. The effect is not a property of the curve or its class. Popcount and bit length of a and b show no association (p ≥ 0.22).
- **No monotone drift.** Speed against position in the session has mean rank correlation −0.10 (19/45 positive), and the decile profile is flat within ±0.5 %.
- **Code.** `rho.plain`'s step is the affine addition $\lambda = (y_2 - y_1)(x_2 - x_1)^{-1}$, which uses neither a nor b.

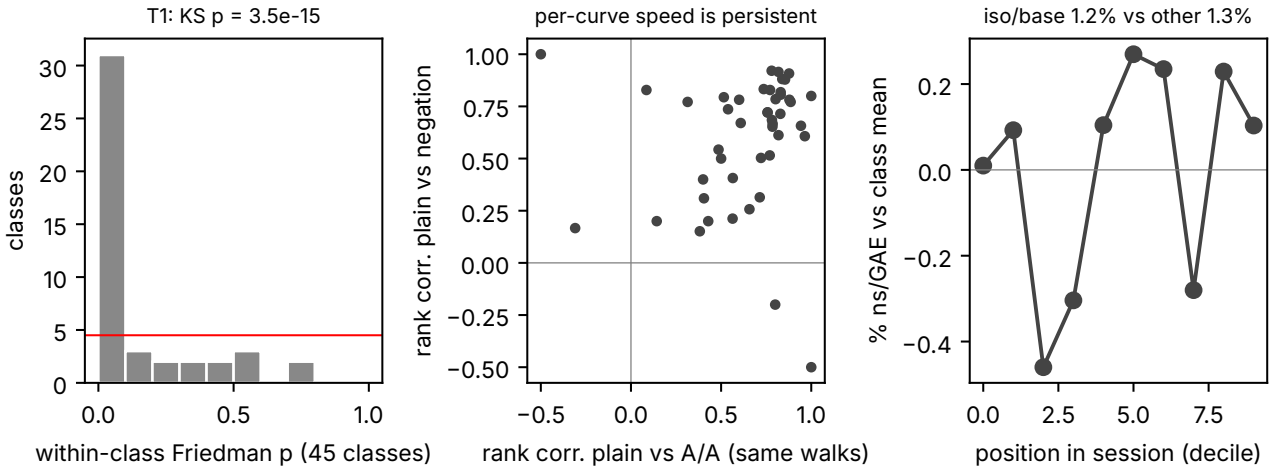


Figure 1: Left: within-class Friedman p-values. Middle: per-curve speed agrees across the A/A and negation arms. Right: no monotone trend with session position. Source: `phase4/analysis.json`, `diag_*.json`.

Reading

Phase 4 cannot separate a representation-dependent cost (the specific coordinates) from non-monotone host fluctuation over each curve's time window. It is **inconclusive** on the question asked. What it bounds: whatever the cause, curves differ by about 1 % SD (at most about 8 % max/min) in measured ns per group operation, and the difference does not follow the isomorphism

class. The required follow-up (PROTOCOL §6) is a preregistered run that interleaves curves: several short sessions per class, each listing the curves in a seeded random order with fresh targets, blocked on (session, target), with no other process on the host.